

Next-Gen AI Compiler

Agents on the Control Plane

Agents own search · orchestration · synthesis.

Compilers own lowering · legality · measure · fallback.

Ask: Is the hybrid prediction (~2027-31) the right bet for your roadmap?

Expert briefing · ~25-30 min + Q&A · SURVEY \$5 primary

Agenda



1. Verdict + four jobs

2. Architecture evolution

3. Prediction checkpoints

4. Commercial blockers (top 5)

5. Gaps that gate the future

6. Trends · stance · ask

Weight

§5 prediction > §4 gaps > §1 trends > rest

Contract

Diagrams & checkpoints first.
We do *not* average conflicts.

Executive verdict

Agents reshape the **control plane**
more than they replace the **data plane**.

Compilers for AI
× AI for compilers



Hybrid LLM-compiler loops



4th job Tier A:
bring-up / codesign

Will not happen soon

- Unconstrained LLM replaces Inductor / opt
- Autonomous chip tape-out via compiler agents (C10)

Four agent jobs — architecture spine

(a) Online

propose →
measure
admit

CompileIQ / GEAK
ACCLAIM / HintPilot

(b) Offline

evolve heuristics
→ C++ / MLGO

Magellan
AlphaEvolve

(c) Eng.

oracle-gated
PR / Change

Archer

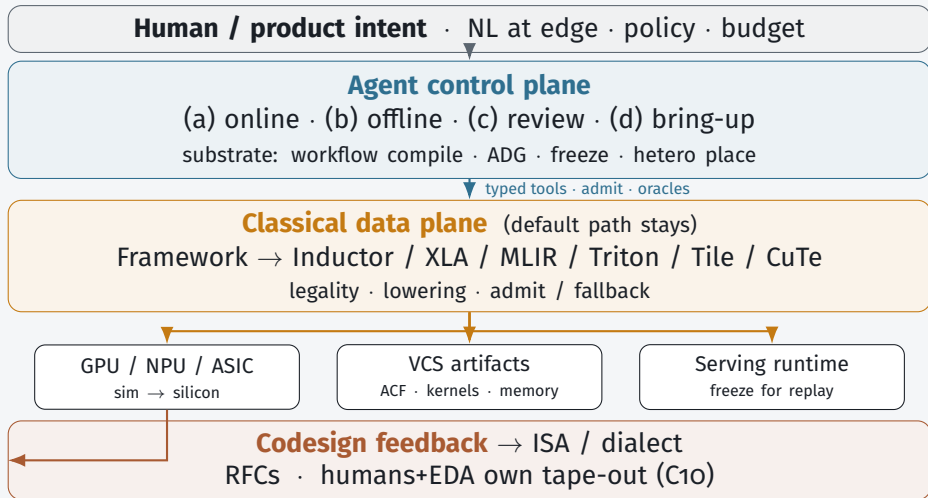
(d) Bring-up

coverage → perf
sim + silicon

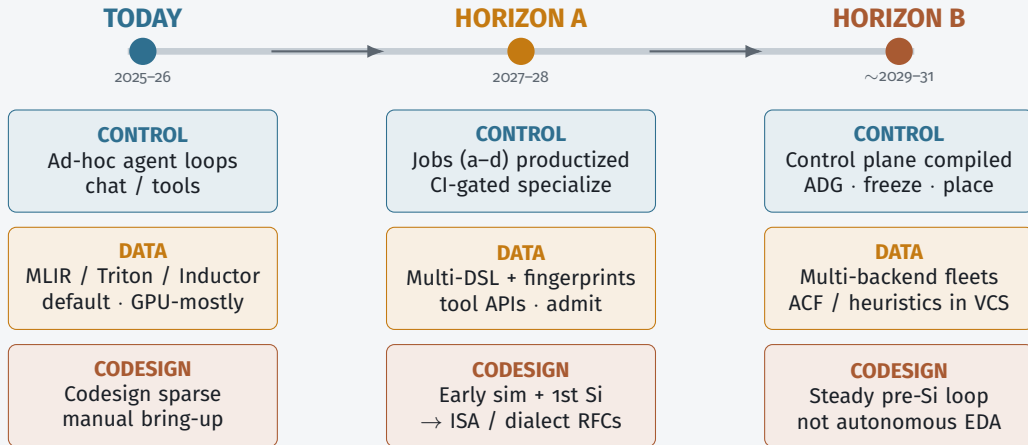
TritorX
KernelEvolve

Control plane jobs sit *above* classical lowering — not instead of it.

Architecture — target stack (§5.1)



Architecture evolution — Today → A → B



Data plane never goes away — agent graph becomes compiled, audited, amortized.

What ships / does not by 2028

Predicted to ship

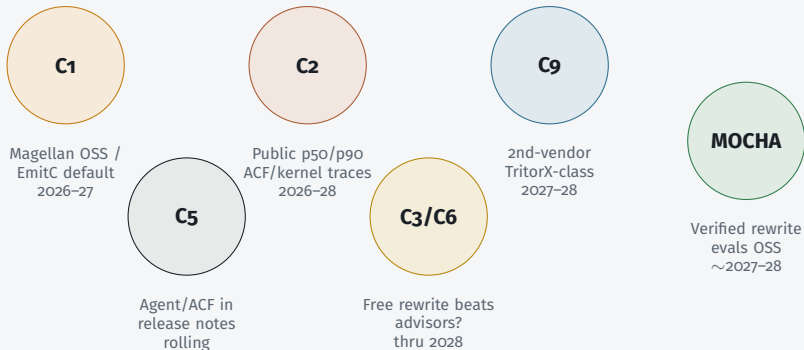
- Agent-addressable tool APIs
- Hot-path specialize (not silent default-all)
- Magellan *and* MLGO both live
- Oracle PR review in serious orgs
- Coverage-first ASIC bring-up
- Triton-family primary; multi-DSL rising

Does not ship

- Unconstrained LLM replaces opt/Inductor
- One agent IR for all vendors
- Kernel agents uniformly beat eager on fusion ladders
- Autonomous microarch tape-out

Horizon A success = CI-gated specialize + oracles + freeze artifacts

Checkpoints — when the prediction changes



Falsifiable sketch on C1–C10. We watch settlement signals — we do not average disagreements.

Checkpoint C1 — heuristics vs neural advisors

A — Magellan path

Evolve shippable C++
(EVOLVE-BLOCK)

Offline coding agent
is the control-plane output

Product = evolutionary coding agent +
review

B — MLGO / EmitC path

NN advisors stay in-tree

June 2026 PoR:

internal inliner →

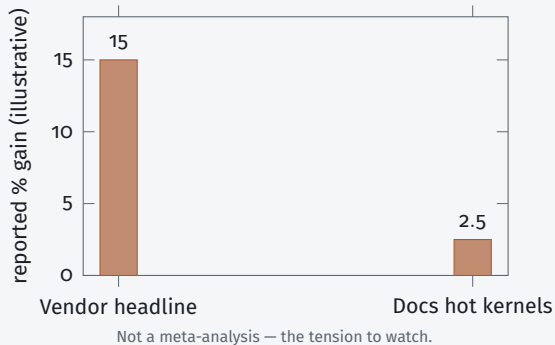
Android/Fuchsia →

Chrome multi-model

Agents train/feature NNs

Settlement: public Magellan llvm patches that dis-
place MLGO or EmitC-MLGO becomes customer default
Watch LLVM syncs + OpenEvolve recipes quarterly through 2027

Checkpoints C2 + C5 — gains & default path



C2 “Agents become default” needs **p50 CI wins**, not best-kernel blogs.

C5 Online specialize (ACF/hints) vs offline eng (Magellan/MLGO) — both may live; *which is the default flag?*

Settlement: pinned public traces + release notes listing agent/ACF workflows.

Checkpoints C₃ / C₆ / C₉ / C₁₀ — API width & scope

C₃ — free rewrite vs advisory

Flip: shared suite where free rewrite wins

⇒ wide rewrite API vs narrow ACF/hints

C₆ — replace vs control plane

Flip: production default with *no* admit
⇒ rewrites survey baseline (now C₆-B)

C₉ — coverage vs peak

Flip: TritorX → KernelEvolve ladder public

⇒ SKU: coverage SLA then perf SLA

C₁₀ — codesign vs auto tape-out

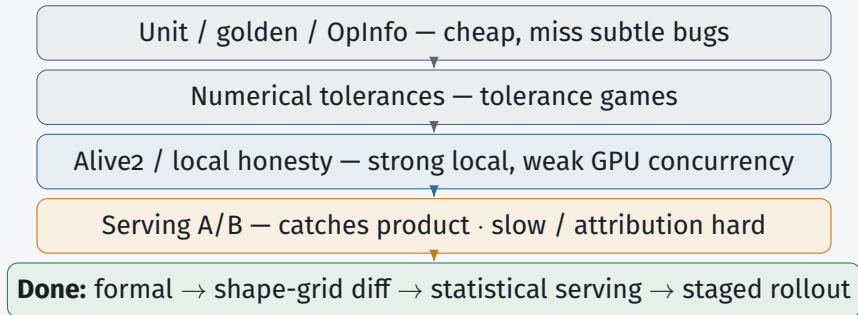
Flip: microarch primarily agent-proposed
and compiler-oracle validated ⇒ enters EDA

Commercial blockers — top 5

- 1 Oracles strong enough for money
Fast-but-wrong · liability
- 2 Cost / replay / when agents run
Nondeterministic \$N compiles
- 3 Agent↔compiler contract
NL-only = demo
- 4 Distributional production evidence
No default-path A/B
- 5 Ownership · supply chain · HITL
Unowned agent code in TCB

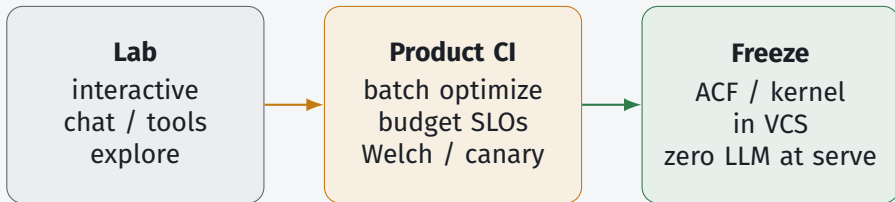
Each blocker gets one slide. Survey lean from \$5.7 + \$4.

Blocker 1 — Oracles for money



Room question: who owns false negatives when admit passes and production miscompiles?

Blocker 2 — Cost, replay, when may the agent run?



Cache key: (IR hash, HW, compiler ver, agent policy) · budget \$/%gain

Falsifies commercially: “chat with compiler on every build” without spend/latency SLOs.

Blocker 3 — Agent↔compiler contract

A. NL + pasted logs — demo only

B. Structured admit traces — CI/SBOM

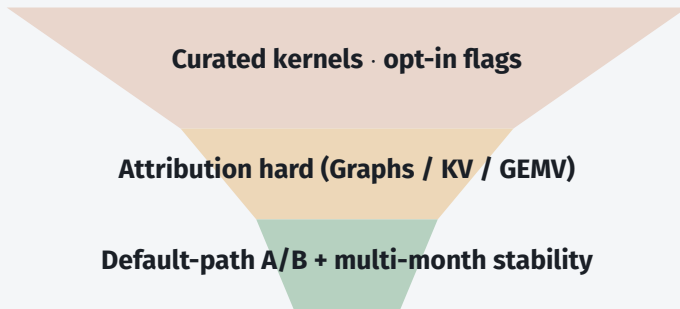
C. Typed tool APIs — required action space

D. Hybrid — NL view over traces

```
admit_record = {graph_hash, hw_id, compiler_ver, action[], oracle[],  
artifact_digest, policy_id}
```

Lean: **C + B**; NL is a view, not source of truth. ACF = Advanced Control File.

Blocker 4 — Distributional production evidence



Marketing bar: p50/p90 + cost-to-compile · MLGO-style persistent QPS, not single-kernel headlines

Blocker 5 — Ownership, security, HITL

TCB risk

agent kernels /
heuristics enter
trusted base

Ownership

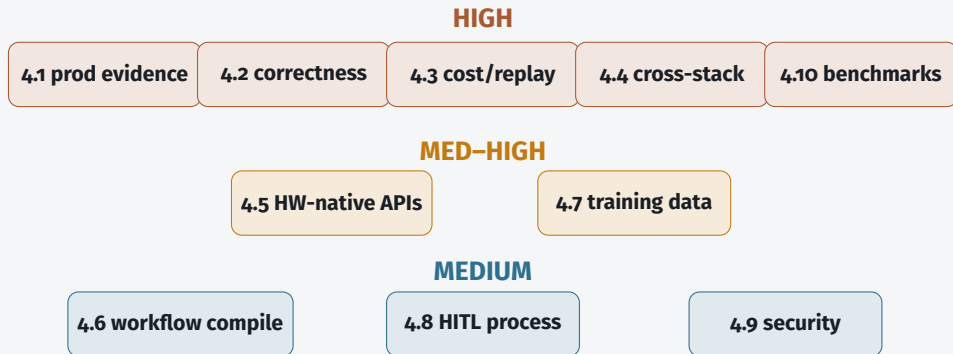
CODEOWNERS
signed provenance
sandbox

HITL capacity

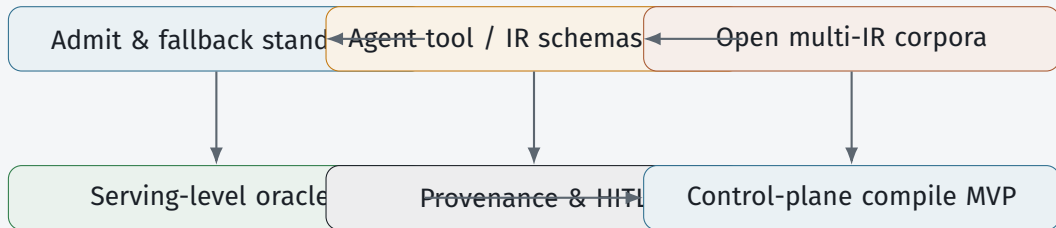
agents $\uparrow$ drafts
reviewers =
bottleneck

Lean: human CODEOWNER + signed admit + sand-
box · oracle auto-merge only for **narrow** action classes
Generic forge AI $\neq$ compiler-oracle review (C7)

Gap map — blockers to §5, not a wishlist



Cross-cutting research agenda



Near-term: specs that make hybrid pipelines CI-regressable.

Org adoption questions

1. Online (CompileIQ) vs offline (Magellan) — which matches your release model?
2. Oracle stack: Alive2 / golden / serving A/B — who owns false negatives?
3. Agent contract surface: LLVM / MLIR / Triton / StableHLO / Tile?
4. How do you cache/regress traces across compiler *and* model upgrades?
5. Max \$/build for a median X% win? Named maintainer per admitted artifact?

Trend backdrop — two stacks converging



Compilers for AI
mature · fragmented substrate

AI for compilers
rapid hybrid growth

Next-gen $\neq$ throw away LLVM/MLIR — make them **agent-addressable**.

Six active trends

A Hybrid guidance

free IR rewrite fragile
constrain action space

B RL → agents

tools · heuristics
workflow-compile

C MLIR + Triton

peak often still
vendor libs / Tile

D Kernel agents

industrial · KernelBench hard

E Verify in loop

local formal $\gg$
whole-program GPU/FP

F Broader object

FMware · mid-decode
agents as eng

Keep substrate, change control plane

Preserve

deterministic lowering
library kernels
CI regressability
local formal checks

Adopt

advisory search
multi-agent orchestration
heuristic synthesis
profile / verifier feedback

**Anti-pattern: replace opt/Inductor
with unconstrained LLM codegen**

Working stance until conflicts settle

- 1 Hybrid control / data plane (C6-B)
- 2 Magellan || MLGO — parallel bets (C1)
- 3 Discount single-number speedups (C2)
- 4 Constrained actions + strong oracles (C3)
- 5 Demote generic SCM AI (C7)
- 6 Coverage→perf ladder; no autonomous EDA (C9/C10)

Claim status snapshot

Supported

A1 planes

A2 four jobs

A3 artifacts

A5 no free replace

S4 ASIC TTM

Contested / Watch

A4 defaults

P1 C1 bet

P2 multi-DSL

S5 profiler APIs

Full claim map: SURVEY §7.

Systems gallery — mechanism clusters

Pass / heuristic

LLM Compiler · Compiler-R1 · Magellan
MLGO · HintPilot · ACCLAIM

Kernel / DSL

GEAK · KernelLLM · AgentCompile
CompileIQ · CuTeGen · AutoKernel ·
KForge

Bring-up / codesign

TritonX · KernelEvolve
Ascend hierarchical diagnosis

Boundary / negative

mlirAgent — below identity
on free IR rewrite

Prefer mechanisms over headline speedups.

Commercial checklist — handout

Architecture

typed tools + admit
traces
VCS $\gg$ dense $\gg$ scratch-
pad
FSM / plan for SLA
freeze before default-on

Trust

layered oracles
CODEOWNERS + prove-
nance
joint version pins
A/B before “prod default”

Business

sell regressable artifacts
+ CI quota
customer owns outputs
coverage then perf SLA
token budget per %gain

Discussion

Thank you — hybrid bet stays the ask

1. Which checkpoint flips your investment first — C1, C2, or C9?
2. Harder commercial bar: **oracle strength** or **replay/freeze**?
3. Building job (a), (b), or (d) first — what do you refuse online?